

Bias Analysis Report

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1. Executive Summary

This report summarizes detected bias signals across data, model behavior, and inferred causal mechanisms.

Key quantitative takeaways:

- Distributional bias: 22 features analyzed; 12 show severe imbalance and 8 moderate divergence.
- High-dimensional bias: 10.0% severe, 15.1% moderate anomaly samples detected.
- Community bias: low=77.8%, severe=7.4%, moderate=14.8% across detected groups.

Detailed interpretation and cross-method consistency are provided in the Overall Fairness Dashboard.

Bias, Fairness & Causal Impact - Executive Dashboard

Bias Summary Scorecard

Dimension	Status	Interpretation
Distributional Bias	High	Feature imbalance vs reference
High-Dimensional Bias	Localized	Population anomalies
Community Bias	Low	Cluster-level imbalance
Algorithmic Bias	Low	Model / error disparities across groups
Group Fairness Gaps	N/A	No sensitive attributes evaluated (fairness not assessed)
Causal Fairness	Material	Estimated causal effect on bias

Interpretation: Green = minimal bias, Orange = moderate imbalance, Red = strong bias risk.

Model Fairness & Causal Impact

Task: regression | Sensitive attributes evaluated: 0

Interpretation note: No explicit sensitive/protected attributes were provided or detected in this run. Therefore, group-based fairness disparities (e.g., MAE/RMSE gaps by subgroup) are not computed, and the fairness section reflects 'not assessed' rather than 'proven fair'.

Fairness alerts triggered: N/A (no sensitive attributes evaluated)

Worst observed fairness gap (summary): N/A (no sensitive attributes evaluated)

Features with significant causal effect on bias: 3

Causal note: Significance is assessed via bootstrap confidence intervals on the estimated effect of a feature on the bias signal (anomaly score). Method: AIPW.

Note: Fairness metrics reflect observed disparities in outcomes/predictions (when sensitive groups are available). Causal analysis (when available) estimates whether changing a feature would change the bias signal under a model-based assumptions framework; it does not, by itself, establish policy or legal discrimination.

Top Bias, Fairness & Causal Drivers

- CRC_incidence | Signals: Dist | Severity: Severe
- diet_high_in_processed_meat | Signals: Dist | Severity: Severe
- diet_high_in_red_meat | Signals: Dist + SHAP | Severity: Severe
- diet_high_in_sodium | Signals: Dist | Severity: Severe
- diet_high_in_sugar_sweetened_beverages | Signals: Dist | Severity: Severe

1b. Statistical Confidence & Robustness

The findings in this report are based on statistical estimates derived from the available dataset and model outputs. While multiple complementary techniques are used to assess fairness and bias, results should be interpreted with the following considerations in mind:

- All metrics are sample-dependent; small subgroups may exhibit higher variance.
- High-dimensional and SHAP-based signals reflect statistical structure, not ground truth.
- Algorithmic fairness metrics capture observed disparities, not guaranteed causal effects.

Confidence is strongest where multiple independent diagnostics (distributional, high-dimensional, and causal) align.

Overall Assessment:

Overall fairness is moderate: feature-level divergence and/or community-linked patterns exist, but evidence does not strongly indicate pervasive structural unfairness under the configured thresholds.

2. Data Profiling Summary

This section provides an overview of dataset completeness and feature composition. The summary below highlights feature types, missingness, and key statistical patterns.

Summary:

- Total Variables: 22
- Numeric Features: 21
- Categorical Features: 1

Profiling Alerts (Top 5):

No missing or data quality issues detected.



Figure: Missingness distribution for top-5 features. This visualization highlights the most incomplete variables.

Key Takeaway:

12 features show strong (severe) distributional divergence and 8 show moderate divergence relative to reference distributions. These signals reflect feature-level imbalance and do not, by themselves, imply discriminatory model behavior unless coupled with group-level or algorithmic effects.

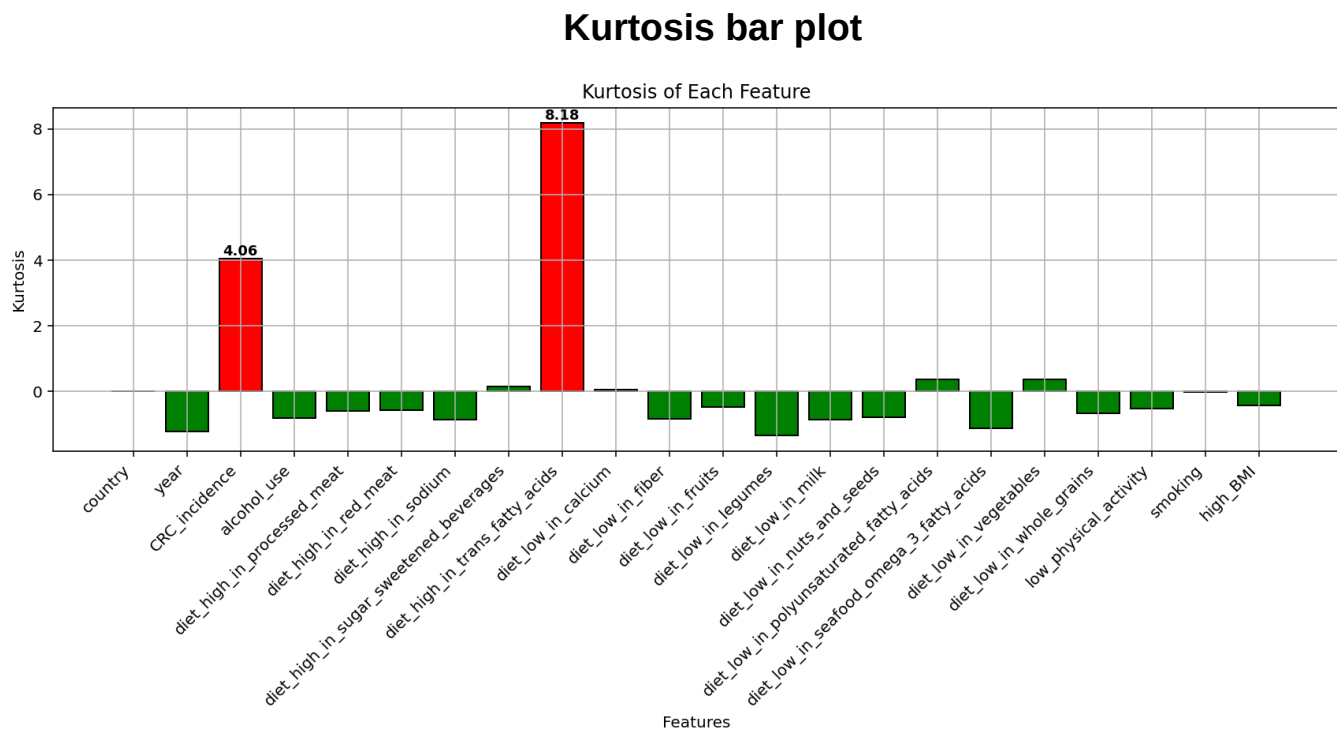


Figure: Kurtosis bar plot. This plot shows the kurtosis values for each feature, highlighting the statistical distribution characteristics.

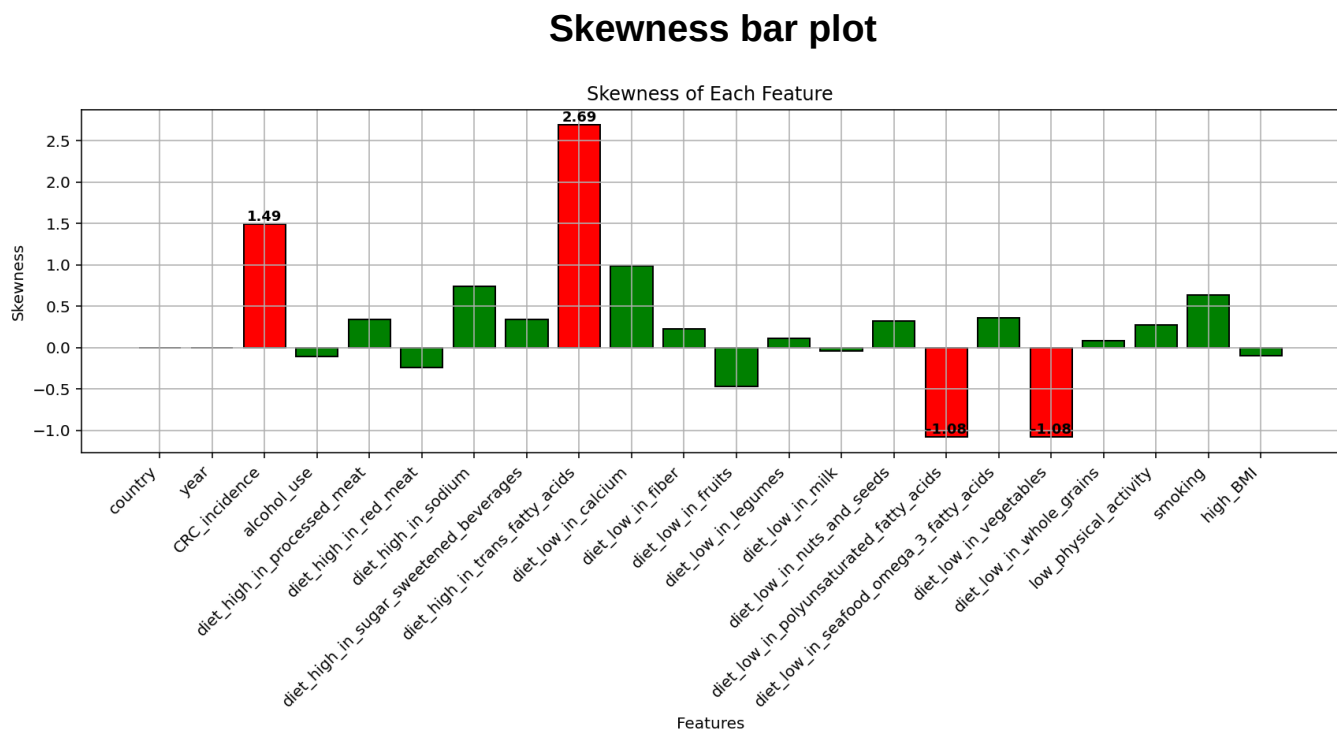


Figure: Skewness bar plot. This plot shows the skewness values for each feature, highlighting the statistical distribution characteristics.

3. Bias Alerts and Key Findings

This section highlights the most biased features in the dataset based on divergence metrics (KL, JS, KS, TVD, and Lp-Norm). Each features alert reflects how much its distribution differs from the expected reference, with severity corresponding to the magnitude of this deviation.

Below are the detailed distributional bias alerts and a compact multi-metric overview.

- Severe bias detected in 'year'. Immediate review recommended.
- Severe bias detected in 'CRC_incidence'. Immediate review recommended.
- Moderate bias detected in 'alcohol_use'. Investigate potential data imbalance.
- Severe bias detected in 'diet_high_in_processed_meat'. Immediate review recommended.
- Severe bias detected in 'diet_high_in_red_meat'. Immediate review recommended.
- Severe bias detected in 'diet_high_in_sodium'. Immediate review recommended.
- Severe bias detected in 'diet_high_in_sugar_sweetened_beverages'. Immediate review recommended.
- Severe bias detected in 'diet_high_in_trans_fatty_acids'. Immediate review recommended.
- Severe bias detected in 'diet_low_in_calcium'. Immediate review recommended.
- Moderate bias detected in 'diet_low_in_fiber'. Investigate potential data imbalance.
- Moderate bias detected in 'diet_low_in_fruits'. Investigate potential data imbalance.
- Low bias detected in 'diet_low_in_legumes'. Monitor for localized issues.
- Moderate bias detected in 'diet_low_in_milk'. Investigate potential data imbalance.
- Severe bias detected in 'diet_low_in_nuts_and_seeds'. Immediate review recommended.
- Severe bias detected in 'diet_low_in_polyunsaturated_fatty_acids'. Immediate review recommended.
- Moderate bias detected in 'diet_low_in_seafood_omega_3_fatty_acids'. Investigate potential data imbalance.
- Severe bias detected in 'diet_low_in_vegetables'. Immediate review recommended.
- Moderate bias detected in 'diet_low_in_whole_grains'. Investigate potential data imbalance.
- Severe bias detected in 'low_physical_activity'. Immediate review recommended.
- Moderate bias detected in 'smoking'. Investigate potential data imbalance.

- Moderate bias detected in 'high_BMI'. Investigate potential data imbalance.

The following figures illustrate how divergence metrics vary across features. Bars exceeding red dashed thresholds represent statistically significant deviations from reference distributions.

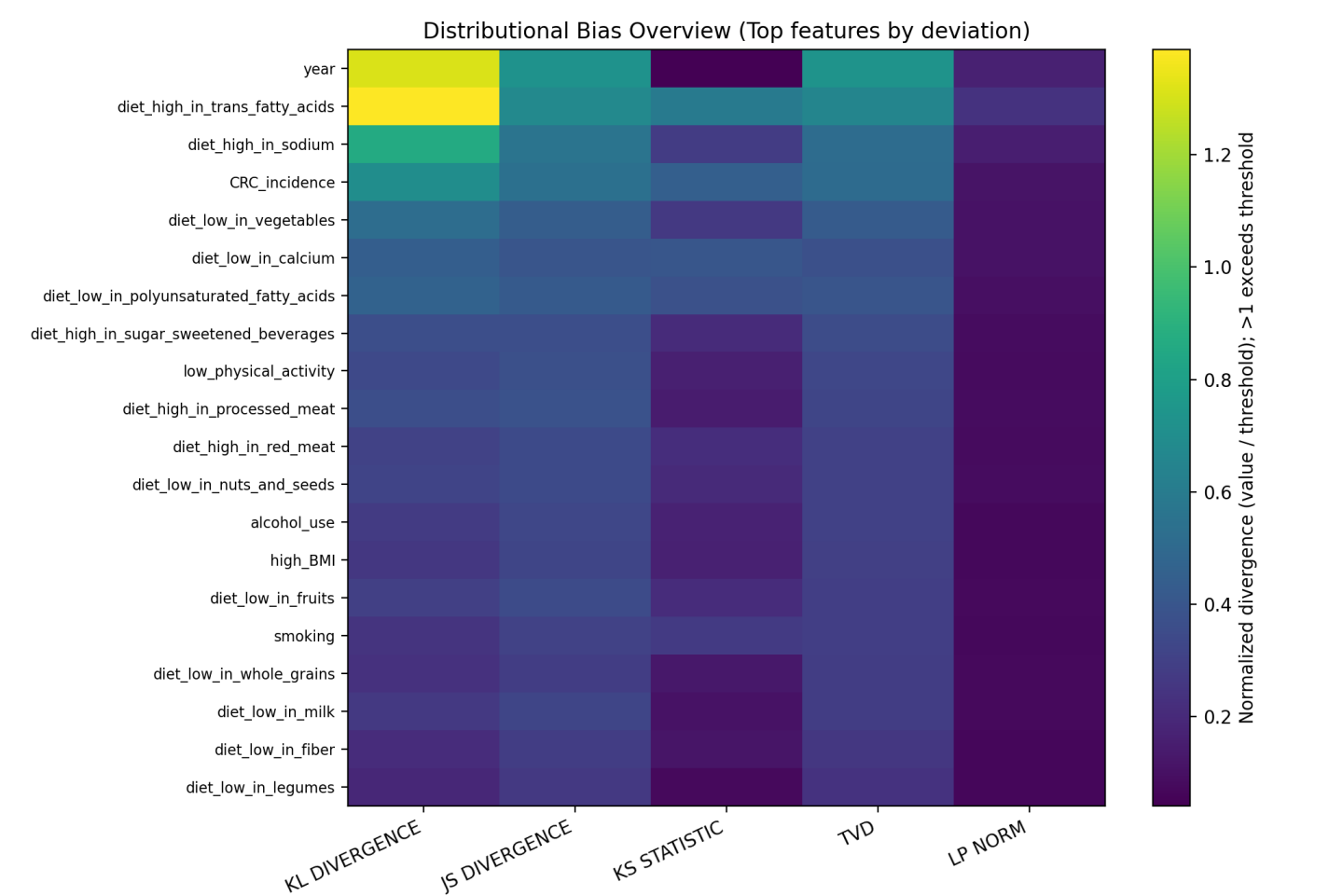


Figure: Combined divergence overview (features × metrics). Cells show divergence normalized by threshold; values > 1 indicate threshold exceedance.

Interpretation Summary: 12 features show strong (severe) distributional divergence and 8 show moderate divergence relative to reference distributions. These signals reflect feature-level imbalance and do not, by themselves, imply discriminatory model behavior unless coupled with group-level or algorithmic effects.

SUMMARY - Features Flagged by 3+ Metrics

- CRC_incidence (Flagged by 5 metrics)
- diet_high_in_sodium (Flagged by 5 metrics)
- diet_high_in_trans_fatty_acids (Flagged by 5 metrics)
- diet_low_in_calcium (Flagged by 5 metrics)
- diet_low_in_vegetables (Flagged by 5 metrics)
- year (Flagged by 4 metrics)
- diet_high_in_processed_meat (Flagged by 4 metrics)
- diet_high_in_red_meat (Flagged by 4 metrics)
- diet_high_in_sugar_sweetened_beverages (Flagged by 4 metrics)
- diet_low_in_nuts_and_seeds (Flagged by 4 metrics)
- diet_low_in_polyunsaturated_fatty_acids (Flagged by 4 metrics)
- low_physical_activity (Flagged by 4 metrics)
- alcohol_use (Flagged by 3 metrics)
- diet_low_in_fiber (Flagged by 3 metrics)
- diet_low_in_fruits (Flagged by 3 metrics)
- diet_low_in_milk (Flagged by 3 metrics)
- diet_low_in_seafood_omega_3_fatty_acids (Flagged by 3 metrics)
- diet_low_in_whole_grains (Flagged by 3 metrics)
- smoking (Flagged by 3 metrics)
- high_BMI (Flagged by 3 metrics)

Top 5 Most Biased Features Per Metric

KL Divergence

- **diet_high_in_trans_fatty_acids (KL Divergence: 1.387)**
- year (KL Divergence: 1.309)
- diet_high_in_sodium (KL Divergence: 0.863)
- CRC_incidence (KL Divergence: 0.701)
- diet_low_in_vegetables (KL Divergence: 0.521)

JS Divergence

- **year (JS Divergence: 0.725)**
- diet_high_in_trans_fatty_acids (JS Divergence: 0.673)
- diet_high_in_sodium (JS Divergence: 0.561)
- CRC_incidence (JS Divergence: 0.536)
- diet_low_in_vegetables (JS Divergence: 0.439)

KS Statistic

- **diet_high_in_trans_fatty_acids (KS Statistic: 0.590)**
- CRC_incidence (KS Statistic: 0.447)
- diet_low_in_calcium (KS Statistic: 0.399)
- diet_low_in_polyunsaturated_fatty_acids (KS Statistic: 0.373)
- diet_high_in_sodium (KS Statistic: 0.283)

TVD

- **year (TVD: 0.730)**
- diet_high_in_trans_fatty_acids (TVD: 0.654)
- diet_high_in_sodium (TVD: 0.517)
- CRC_incidence (TVD: 0.510)
- diet_low_in_vegetables (TVD: 0.429)

Lp-Norm

- **diet_high_in_trans_fatty_acids (Lp-Norm: 0.237)**

- year (Lp-Norm: 0.164)

- diet_high_in_sodium (Lp-Norm: 0.152)

- CRC_incidence (Lp-Norm: 0.114)

- diet_low_in_vegetables (Lp-Norm: 0.109)

4. High-Dimensional Bias Analysis

Severity Distribution Summary:

Total samples assessed: 729
severe: 10.0% | moderate: 15.1% | low: 74.9%

Interpretation: Higher severe percentages suggest stronger concentration of atypical or imbalanced samples in high-dimensional feature space.

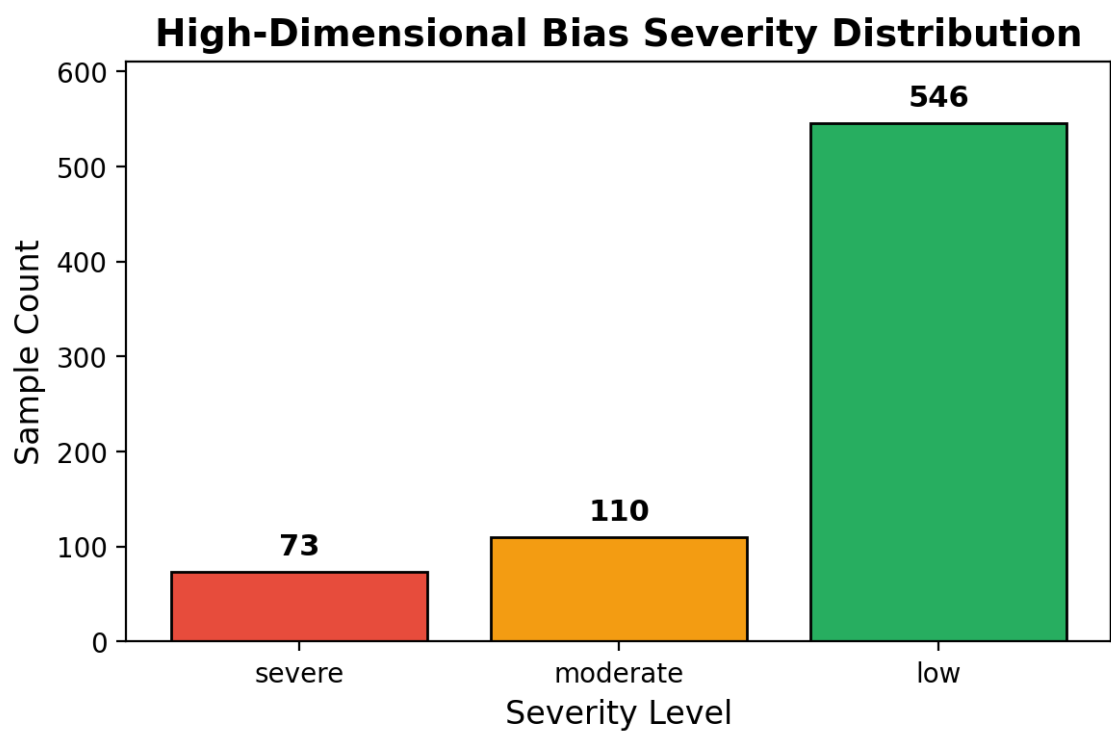


Figure: High-Dimensional Bias Severity Distribution. Bars indicate the share of samples classified as Severe, Moderate, or Low bias.

LOF Score Distribution

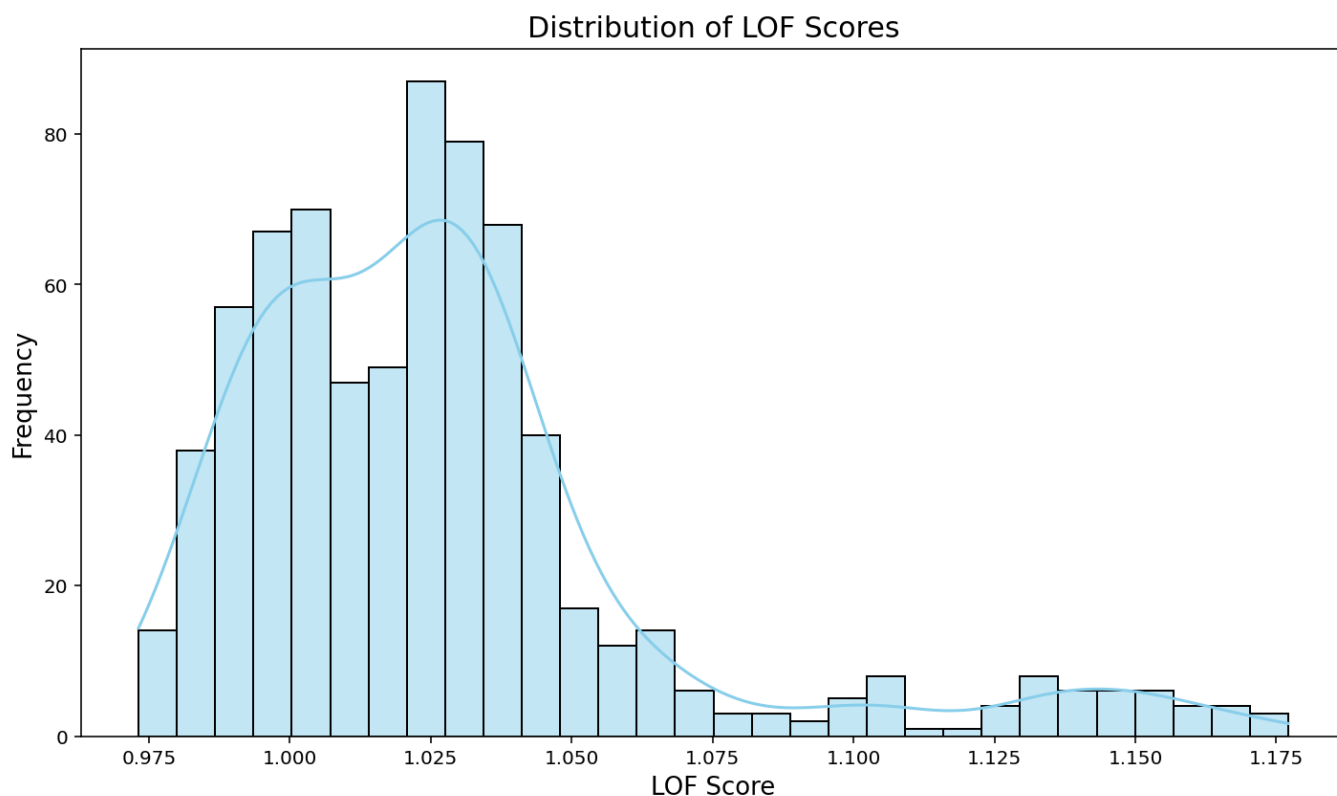


Figure: Distribution of LOF anomaly scores. Right-tail samples correspond to stronger bias signals.

Isolation Forest Score Distribution

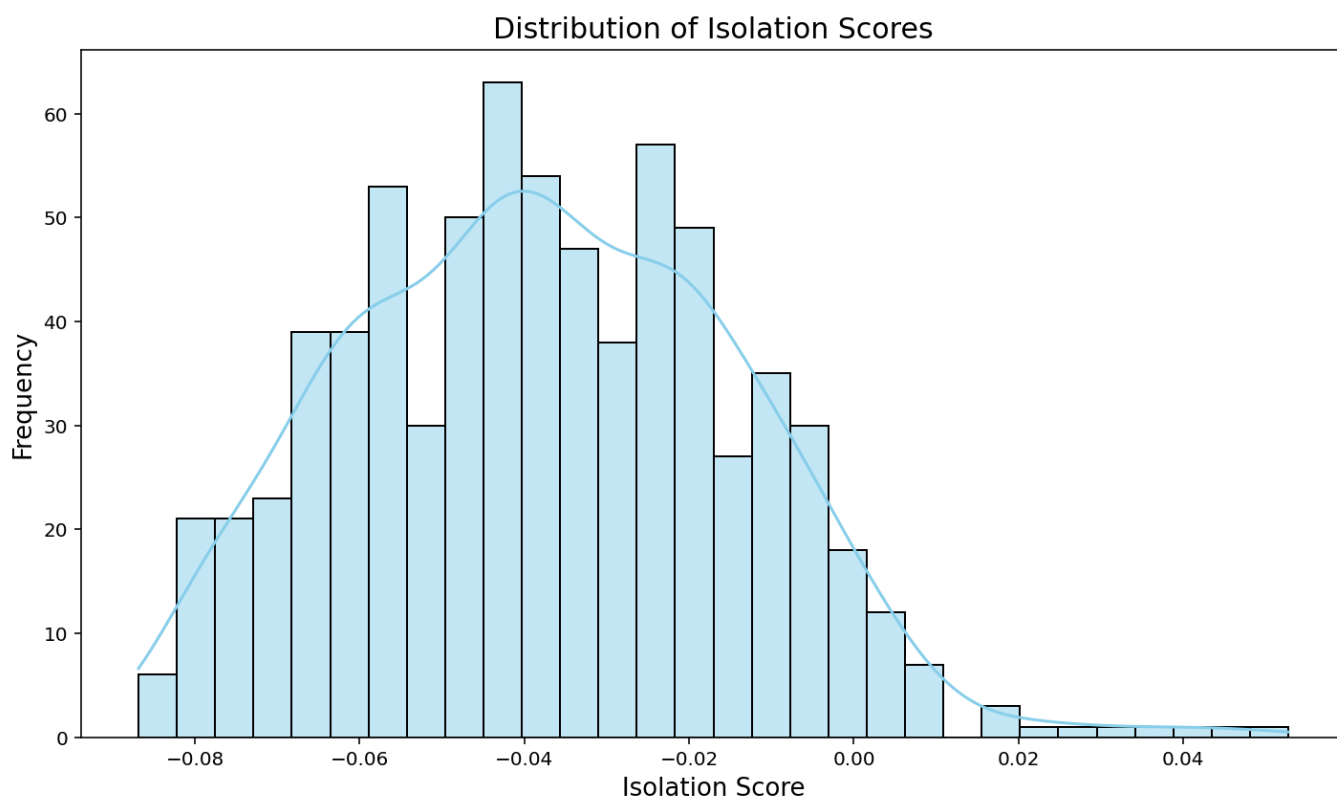


Figure: Distribution of Isolation Forest anomaly scores. Right-tail samples correspond to stronger bias signals.

GMM Score Distribution

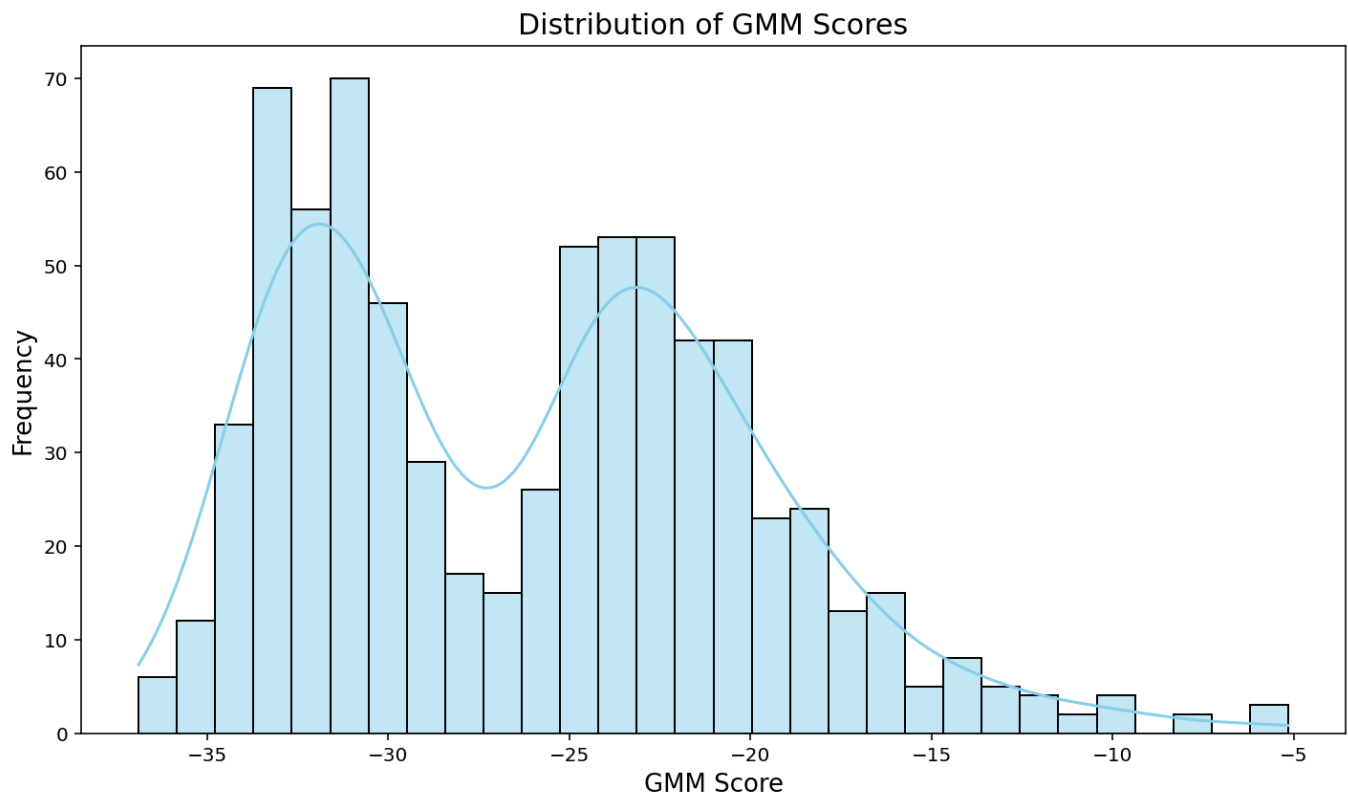


Figure: Distribution of GMM anomaly scores. Right-tail samples correspond to stronger bias signals.

Autoencoder Score Distribution

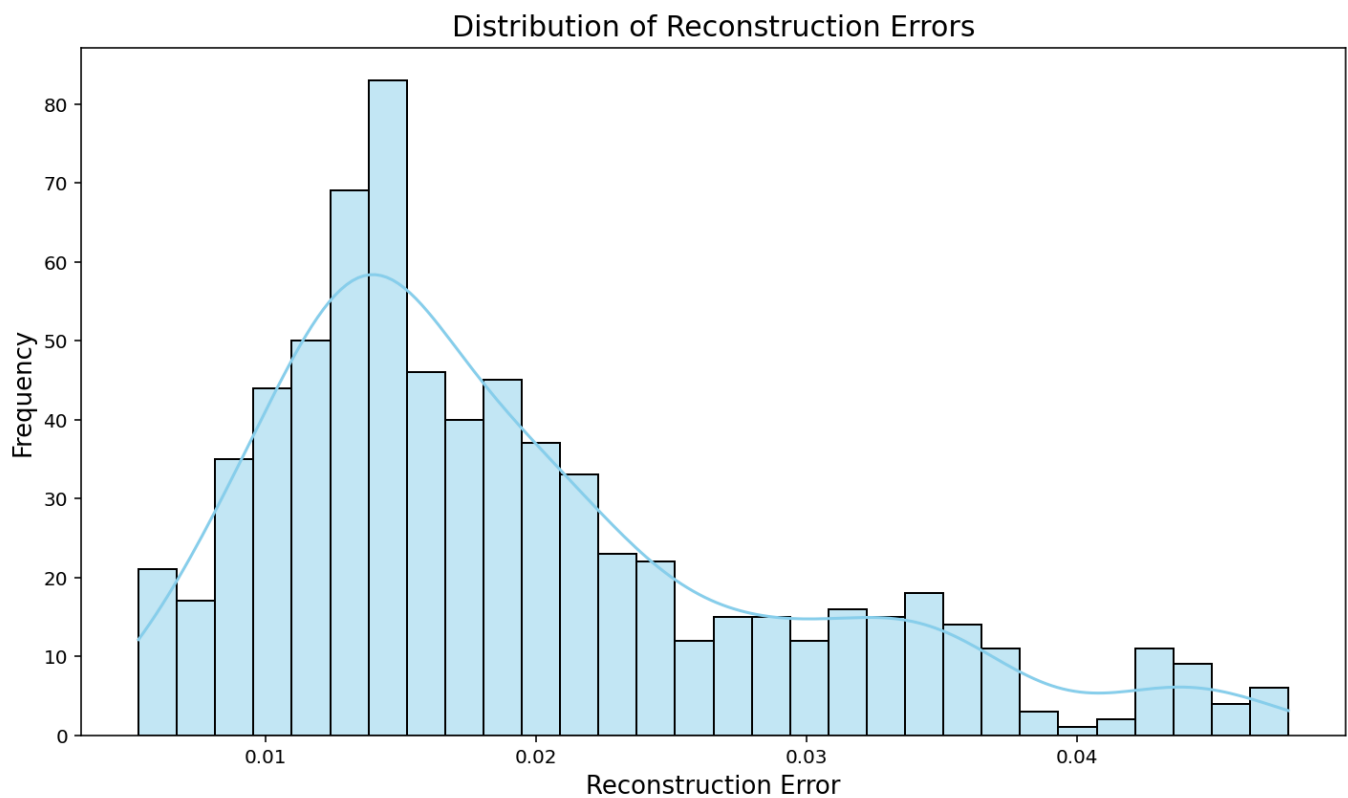


Figure: Distribution of Autoencoder anomaly scores. Right-tail samples correspond to stronger bias signals.

Key Takeaway:

High-dimensional anomaly detection identified 10.0% severe and 15.1% moderate anomaly samples. This

indicates non-uniform bias patterns concentrated in specific regions of the feature space rather than widespread structural imbalance.

Confidence Note: High-dimensional bias scores are derived from ensemble anomaly detection methods (autoencoder, isolation-based, and density-based models). Agreement across methods increases confidence, while isolated signals should be interpreted cautiously, particularly for sparse regions of the feature space.

5. Community Bias Detection

Community bias analysis explores bias patterns across automatically discovered subgroups of samples (communities). These groups are detected through unsupervised clustering in high-dimensional space. Each community is then evaluated for internal imbalance, anomaly concentration, or divergence from global distributions.

Summary:

- Total communities detected: 27
- Flagged as potentially biased (based on thresholds): 6
- Non-flagged (below significance threshold): 21
- Smallest group size: 27
- Largest group size: 27
- HD Community Link (R2): 0.741

Severity Distribution Across Communities:

- low: 77.8% | severe: 7.4% | moderate: 14.8%

Interpretation: While few communities crossed the hard flagging threshold, quantile-based severity analysis shows varying bias intensity - with certain groups exhibiting stronger deviations (Severe / Moderate / Low). This helps identify which clusters warrant closer inspection for systemic imbalance.

Community Detection from Similarity Graph

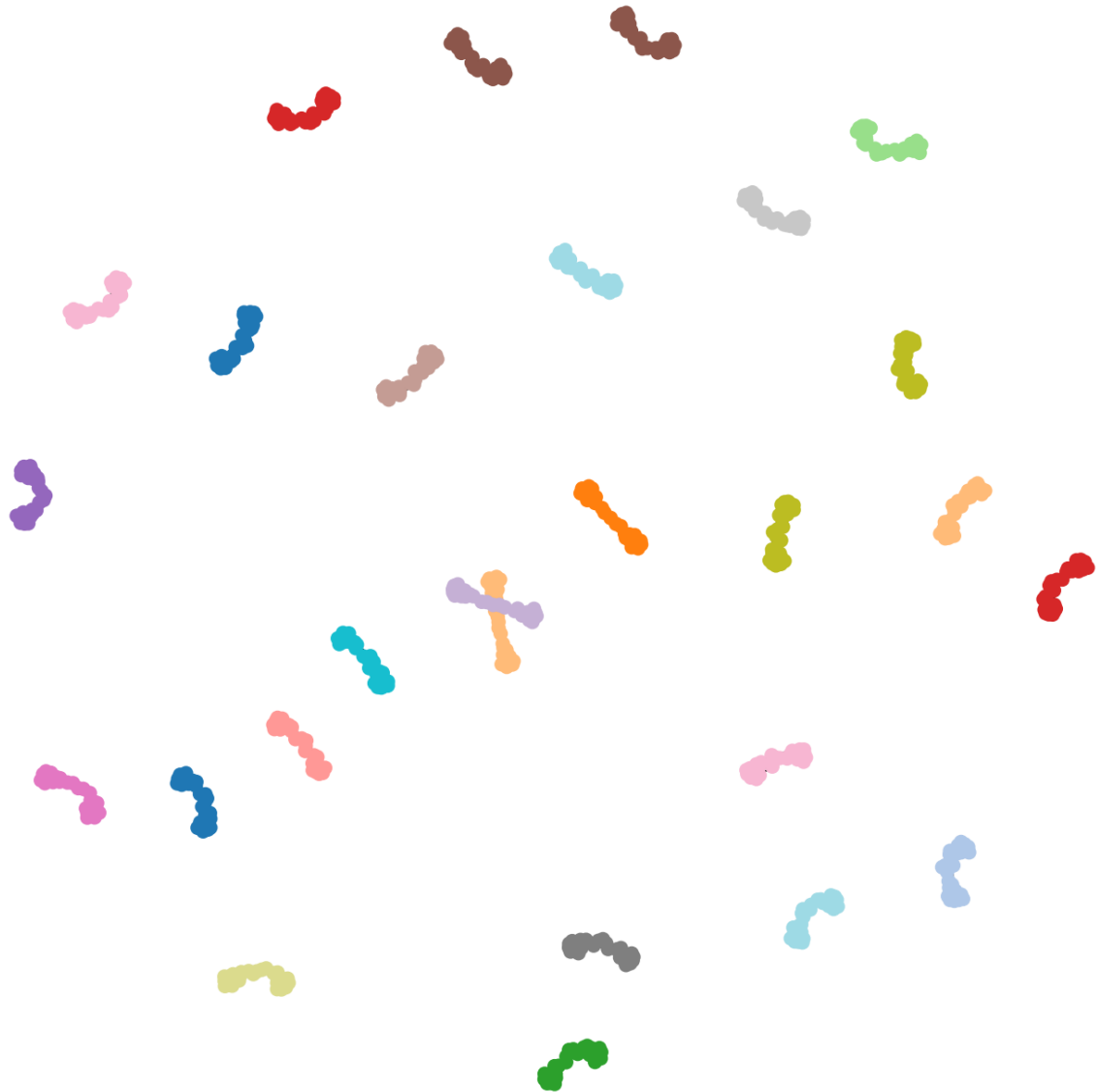


Figure: Community structure network graph showing detected subgroups and their interconnections. Denser nodes correspond to closely related samples.

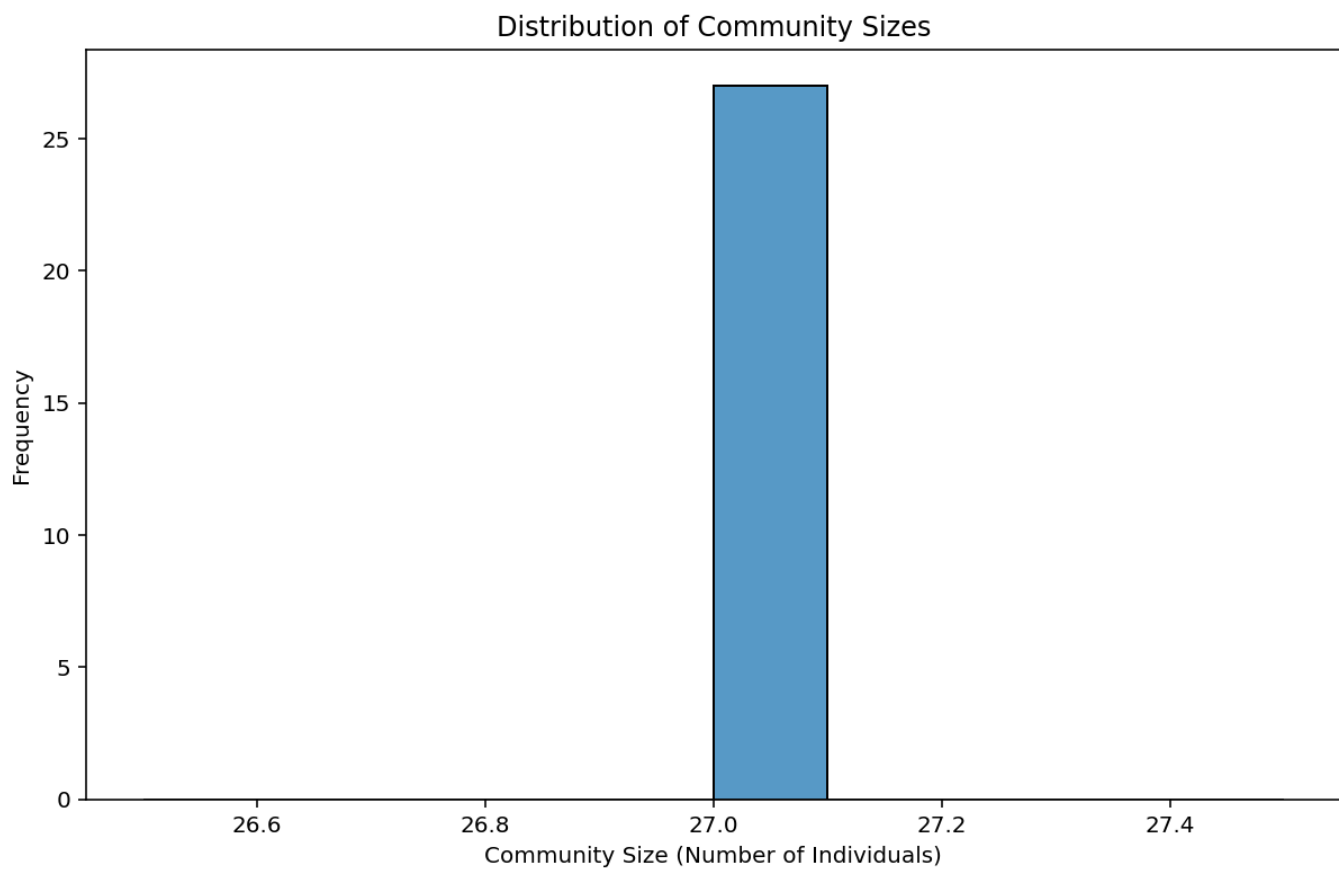


Figure: Size distribution of detected communities. Larger groups may dominate overall patterns, while smaller ones can indicate underrepresented segments.

Community Bias Severity Distribution

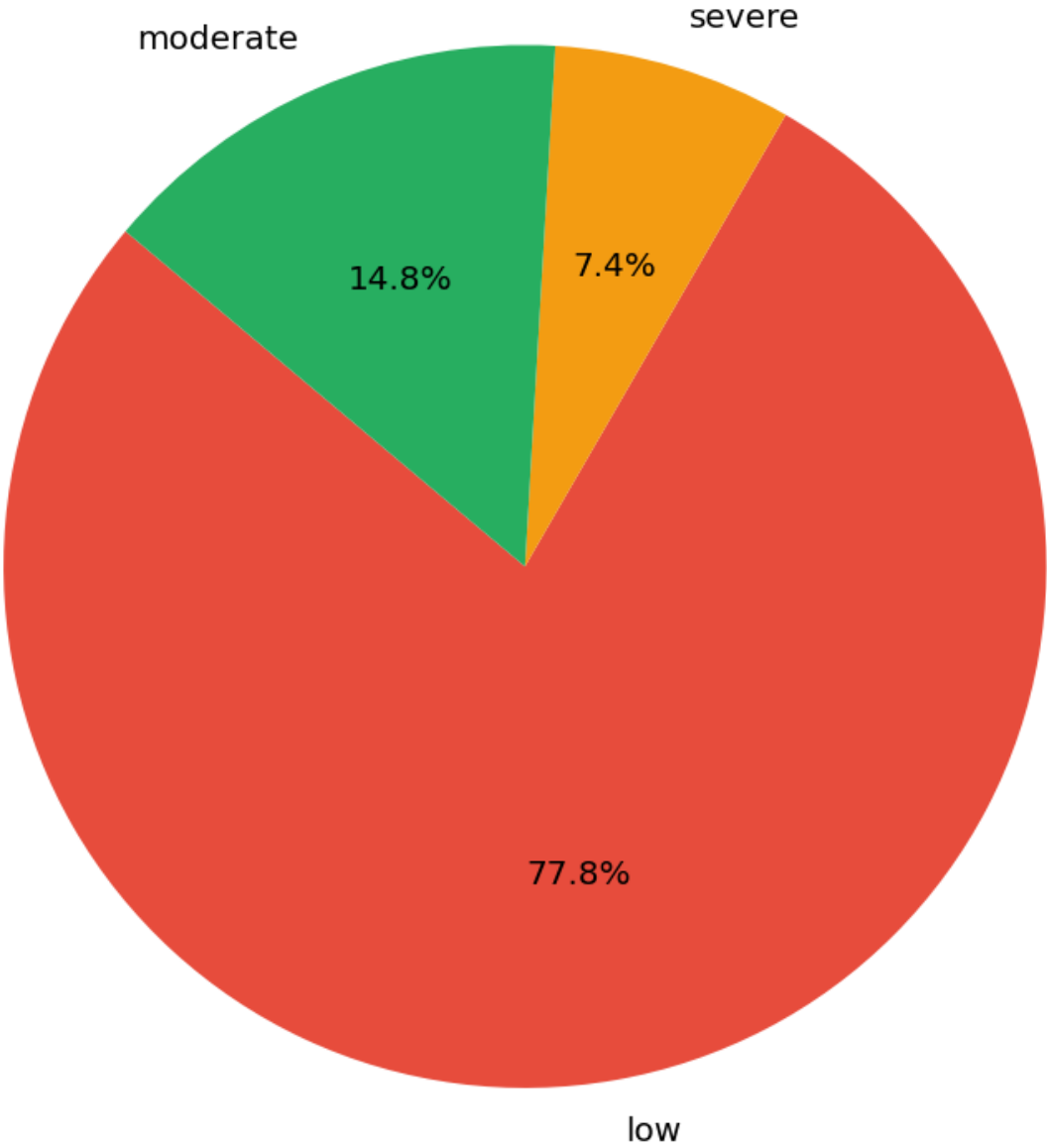


Figure: Community Bias Severity Distribution - showing proportions of Severe, Moderate, and Low bias communities, derived from mean anomaly scores.

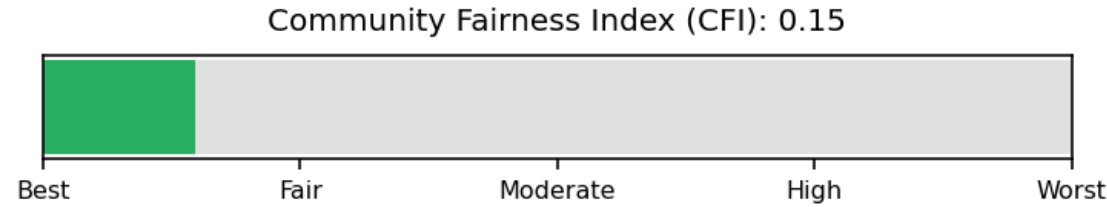


Figure: Community Fairness Index (CFI). A lower CFI indicates greater fairness across communities (balanced severity levels). Values above 0.4 suggest potentially uneven bias distribution among groups.

The computed CFI indicates excellent community-level fairness, with bias evenly distributed and no dominant subgroup effects detected.

Actionability:

Communities flagged with severe or moderate bias represent candidate groups for targeted investigation. Mitigation strategies may include feature re-weighting, data augmentation for underrepresented segments, or community-specific model evaluation.

Key Takeaway:

Overall community-level fairness is strong and balanced (low CFI). Community clustering strongly aligns with bias intensity (high R2), suggesting structural segmentation.

Overall, community-level analysis complements sample-level metrics by revealing whether bias signals cluster within specific population subgroups. These insights help determine if observed biases are localized or structural.

6. SHAP-Based Feature Importance

This section identifies which features contribute most to potential bias using SHAP (SHapley Additive exPlanations). SHAP values quantify how much each feature influences a models predictions - higher mean SHAP values indicate stronger model influence. Features that are both highly influential (SHAP) and repeatedly bias-flagged should be prioritized.

Interpretability Note: SHAP values represent average marginal contributions under model-specific assumptions. They provide insight into model behavior but do not imply causal influence or fairness guarantees.

Interpretability note: model-based TreeSHAP could not be computed for this run. The report therefore uses SHAP-like feature attributions derived from anomaly/reconstruction contributions (method: autoencoder_reconstruction).

SHAP Severity Distribution:

Total features assessed: 10
severe: 1 (10.0%) | moderate: 2 (20.0%) | low: 7 (70.0%)

Interpretation: Moderate SHAP severity indicates some uneven feature influence, though most remain within normal ranges.

The figures below show SHAP-based feature importance and dependency patterns.

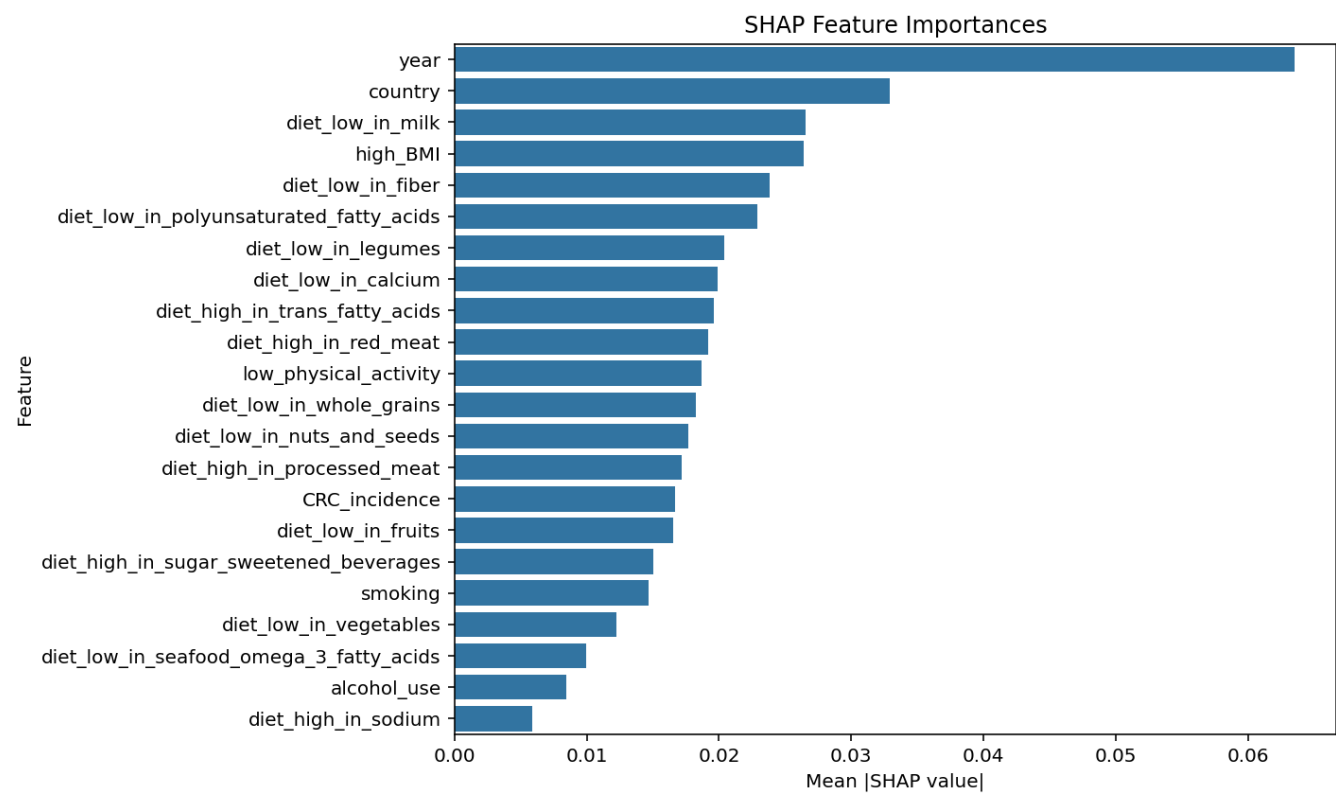


Figure: SHAP Feature Importance - illustrating how feature contributions vary across samples. Longer bars or stronger colors indicate greater impact on model predictions.

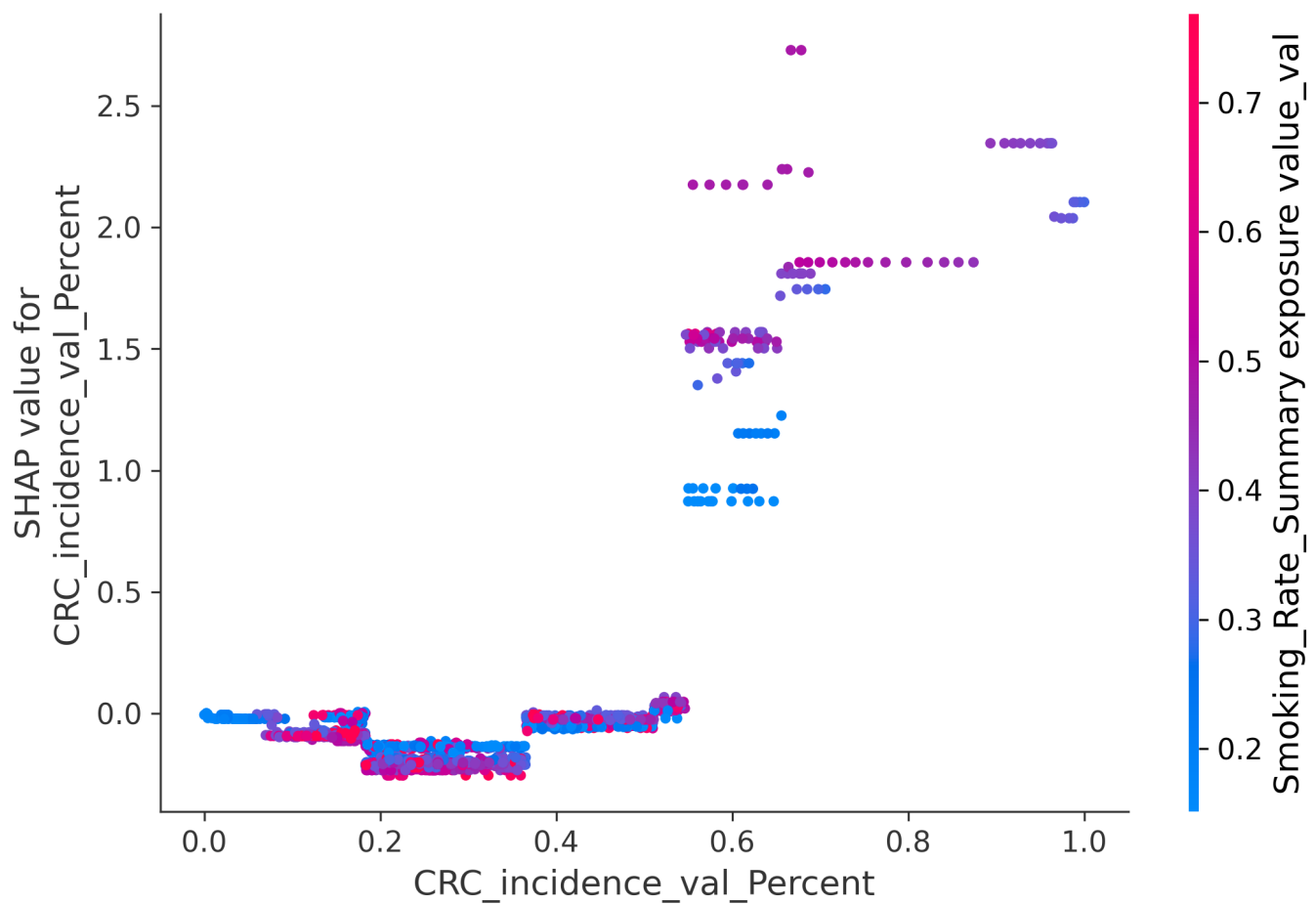


Figure: SHAP Dependence - illustrating how feature contributions vary across samples. Longer bars or stronger colors indicate greater impact on model predictions.

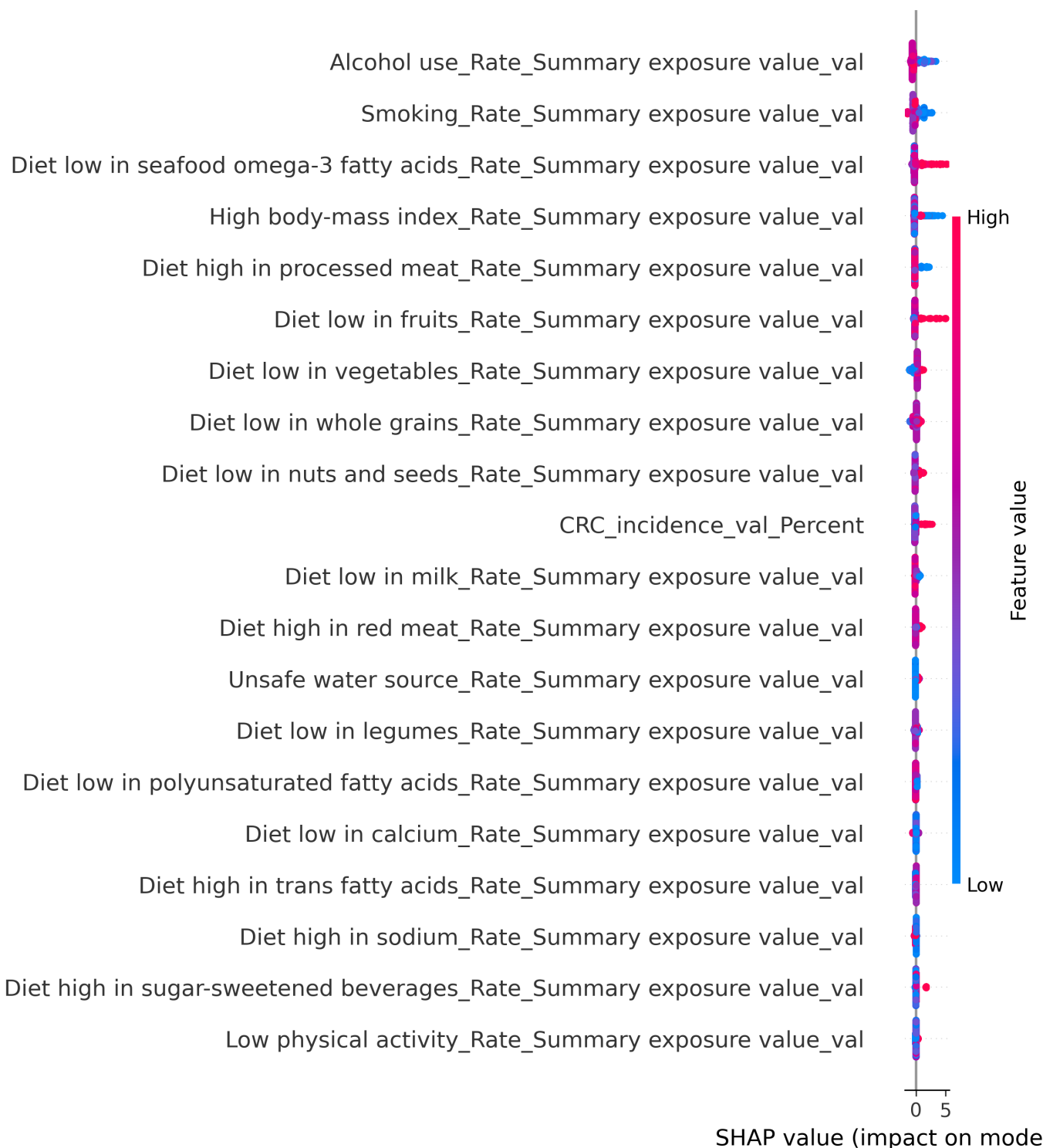


Figure: SHAP Summary - illustrating how feature contributions vary across samples. Longer bars or stronger colors indicate greater impact on model predictions.

SHAP Summary Insights

- SHAP summary metrics were unavailable for this run.

Key Takeaway:

SHAP analysis (where available) provides diagnostic insight into model sensitivity and feature influence. No features exceeded predefined SHAP-based bias thresholds in this run; therefore, no dominant model-level bias drivers were identified.

Interpretation Summary: SHAP analysis (where available) provides diagnostic insight into model sensitivity and feature influence. No features exceeded predefined SHAP-based bias thresholds in this run; therefore, no dominant model-level bias drivers were identified.

Unified Feature Attribution (HD SHAP Community)

This table merges three perspectives:

- High-dimensional anomaly severity
- SHAP-based model influence
- Community-level deviation patterns

Features appearing in many biased communities AND with strong SHAP values should be prioritized for fairness mitigation.

Feature	Mean SHAP	SHAP Severity	# Comms	# Severe Comms	Max	Mean
year	0.0635	severe	0.0	0.0	0.0000	0.0000
country	0.0329	moderate	8.0	0.0	0.5000	0.4183
diet_low_in_milk	0.0265	moderate	4.0	0.0	0.5070	0.4082
high_BMI	0.0264	low	0.0	0.0	0.0000	0.0000
diet_low_in_fiber	0.0238	low	3.0	0.0	0.4379	0.3797
diet_low_in_polyunsaturated_fatty_acids	0.0229	low	3.0	1.0	0.4939	0.4638
diet_low_in_legumes	0.0204	low	12.0	0.0	0.4843	0.3794
diet_low_in_calcium	0.0199	low	3.0	0.0	0.6773	0.4830
diet_high_in_trans_fatty_acids	0.0196	low	1.0	0.0	0.7431	0.7431
diet_high_in_red_meat	0.0192	low	4.0	0.0	0.4213	0.3538

Interpretation:

- High SHAP + many severe communities strong candidate for fairness remediation.
- High Delta but low SHAP community-specific imbalance.
- High SHAP but low Delta global driver but not community-localized bias.

7. Global Bias Radar Overview

Global Bias Radar not available.

Global Stability Index unavailable.

8. Algorithmic Fairness Assessment

This section evaluates the fairness of the predictive model for a regression task. Fairness is assessed by comparing error behavior across sensitive groups (e.g., MAE/RMSE gaps), and by reviewing residual fairness, counterfactual stability, and causal sensitivity where available.

Confidence Note: Algorithmic fairness metrics are computed on observed predictions and reflect empirical disparities rather than guaranteed causal effects. Residual and counterfactual analyses increase robustness but remain sensitive to model specification and feature encoding.

Model: xgb_5Y.pkl

No algorithmic fairness alerts were generated. This reflects either balanced error behavior across evaluated groups or the absence of sensitive attributes in this run.

Residual Fairness Assessment

This section examines fairness in model residuals (errors) across sensitive groups. It highlights whether the model systematically over- or under-predicts for certain groups, and measures error disparities using MAE, RMSE, and (where available) R2.

Global Residual Metrics

MAE:	0.3862
RMSE:	0.7194
task_type:	regression
R2:	0.9955

Residual Fairness by Sensitive Attribute

Attribute: diet_high_in_sugar_sweetened_beverages

No group-level residual fairness metrics were available for this attribute.

Interpretation:

Residual fairness highlights whether prediction errors are evenly distributed across groups. Higher error gaps indicate potential over- or underestimation bias. Low severity means residuals are comparable, while Moderate or Severe alerts suggest biased error patterns requiring closer review.

Summary Insight:

Overall residual fairness appears balanced with no significant disparities.

Global Performance (Regression)

MAE:	0.38619148268500597
RMSE:	0.7193659789828614
R2:	0.9955495861398449

9. Recommended Next Steps

Recommended Actions

- Prioritize review of features with both high SHAP impact and repeated bias flags.
- Investigate communities with severe bias severity for potential underrepresentation.
- Validate model behavior on subgroups contributing most to high-dimensional anomalies.
- Monitor residual and counterfactual fairness metrics during future retraining cycles.

Final Note on Confidence: Bias assessment is an iterative process. Confidence in fairness outcomes increases when assessments are repeated over time, validated on new data, and combined with domain expertise. The results herein should be used to guide investigation and mitigation, not as a one-time certification.

10. Overall Fairness Dashboard

This dashboard consolidates all fairness diagnostics - from feature-level divergence to community bias and SHAP-based model influence - into one holistic summary. It provides a unified view of current fairness conditions across analytical dimensions.

Bias Summary Scorecard

Dimension	Status	Notes
Distributional Bias	High	21 features flagged - 12 severe, 8 moderate, 1 low
High-Dimensional Bias	Localized	73 severe, 110 moderate
SHAP Feature Impact	Present	Top drivers identified
Community Bias	Localized (Fair)	6 of 27 flagged (7.4% severe, 14.8% moderate) CSV=N/A, CFI=0.148, R2=0.741

Interpretation: Green = minimal bias, Orange = moderate imbalance, Red = strong bias risk.

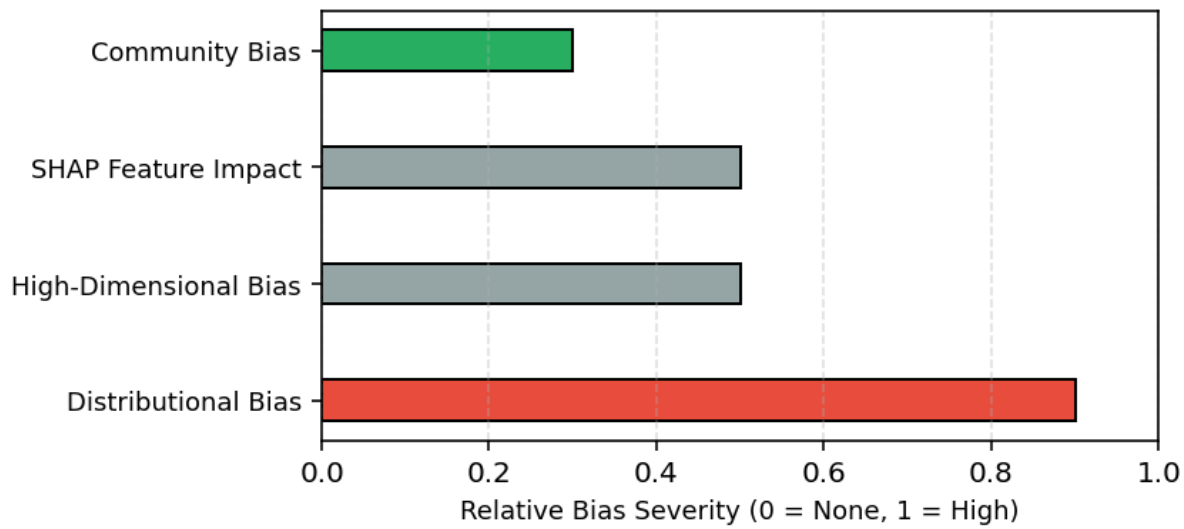


Figure: Relative bias severity per analytical dimension.

Overall Fairness Assessment

Overall fairness is moderate: feature-level divergence and/or community-linked patterns exist, but evidence does not strongly indicate pervasive structural unfairness under the configured thresholds.

Interpretation Guide: Green = minimal bias | Orange = moderate imbalance | Red = high bias risk. Community fairness (CFI) near 0 indicates strong equity; values >0.4 may require subgroup-level review.

11. Causal Fairness Diagnostics

This appendix estimates whether changing a feature would causally affect the bias outcome. We use a doubly-robust estimator (AIPW) with bootstrap to estimate the Average Treatment Effect (ATE).

Method: Doubly Robust AIPW (bootstrap 50). Treatment = feature > median. Outcome = standardized combined anomaly score.

Tested features: year, country, diet_low_in_milk, high_BMI, diet_low_in_fiber

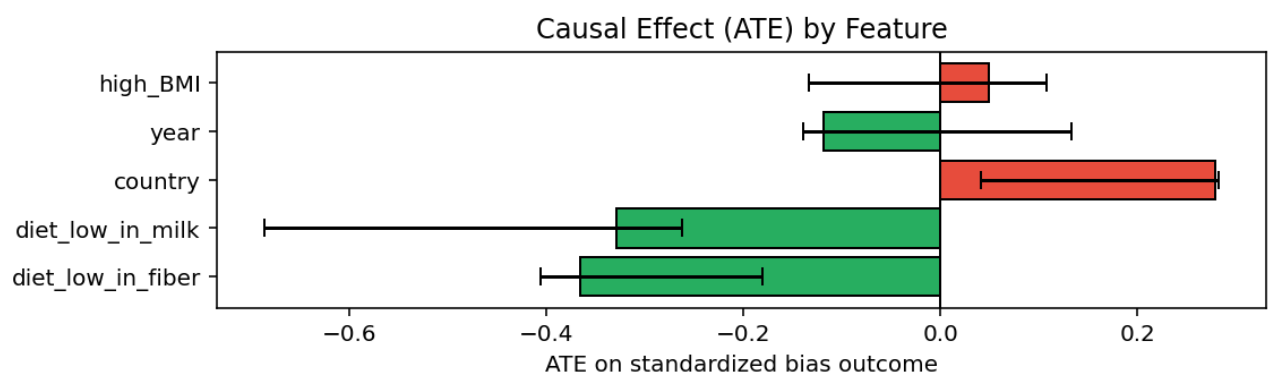


Figure: Estimated causal effect (ATE) of each tested feature on the standardized bias outcome. Error bars show 95% bootstrap CIs.

Positive ATE values indicate that increasing a feature tends to raise overall bias intensity, while negative values suggest the feature mitigates bias signals. The magnitude reflects standardized effect size (in bias-score units).

Top Effects (absolute ATE):

- diet_low_in_fiber - ATE: -0.3663 (95% CI: -0.4061 - -0.1808; threshold=0.4652)
- diet_low_in_milk - ATE: -0.3289 (95% CI: -0.6867 - -0.3952; threshold=0.5042)
- country - ATE: +0.2795 (95% CI: 0.0414 - 0.2828; threshold=0.5)
- year - ATE: -0.1188 (95% CI: -0.1393 - 0.1331; threshold=0.5)
- high_BMI - ATE: +0.0488 (95% CI: -0.1336 - 0.1081; threshold=0.5091)

Interpretation: An ATE above zero suggests that moving the feature from below to above the threshold causally increases high-dimensional bias signals. Negative ATE suggests a protective effect. Results should be read as directional evidence, not definitive causation, especially when features are strongly correlated.

12. Annex 1 - Explanation of Metrics

Kullback-Leibler (KL) Divergence

Measures how one probability distribution diverges from a second reference distribution. High KL indicates significant divergence, signaling potential bias.

Jensen-Shannon (JS) Divergence

A symmetric form of KL divergence that provides a bounded measure (0 to 1) of divergence between distributions. Values closer to 1 indicate significant distributional bias.

Kolmogorov-Smirnov (KS) Statistic

Quantifies the maximum difference between two cumulative distributions, identifying potential distributional biases or shifts.

Total Variation Distance (TVD)

Measures the maximum possible difference between probabilities assigned by two distributions, directly highlighting biased differences.

Lp-Norm

General measure of distance between distributions, where higher values suggest larger divergence or bias in feature distributions.

Autoencoder Reconstruction Error

Quantifies anomalies by measuring how well an autoencoder reconstructs data points. Higher error suggests unusual patterns, possibly indicating bias.

Isolation Forest Score

Assigns anomaly scores based on how easily data points can be isolated; higher scores may indicate unusual distributions or biased patterns.

Gaussian Mixture Model (GMM) Score

Uses probability distributions to model data points. Low likelihood scores highlight unusual data points or potential biases.

SHAP (SHapley Additive exPlanations) Values

Interpret feature importance in model predictions. High SHAP values indicate features significantly contributing to bias or anomalies.

Generated automatically by APPO v2.0 - Automated Fairness Assessment Framework.